INCH-POUND

MIL-S-24187/6A(SH)
22 March 1991

SUPERSEDING
MIL-S-24187/6(SH)
15 June 1978

MILITARY SPECIFICATION SHEET

SWITCH ASSEMBLIES, LINEAR MOVEMENT, MANUAL AND REMOTE OPERATION, SNAP-ACTION, STYLE SLS

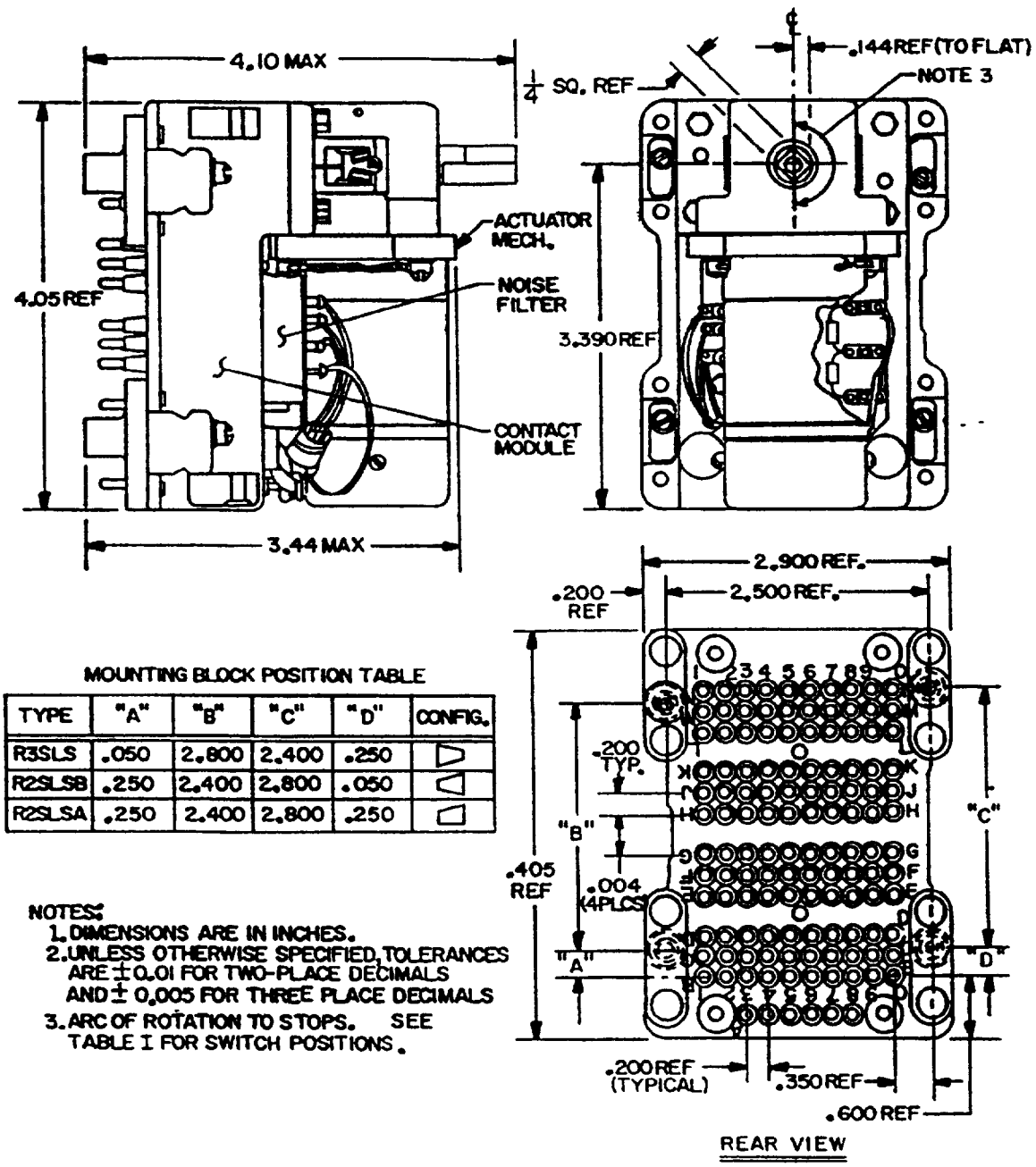
This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-24187.

AMSC N/A FSC 5930

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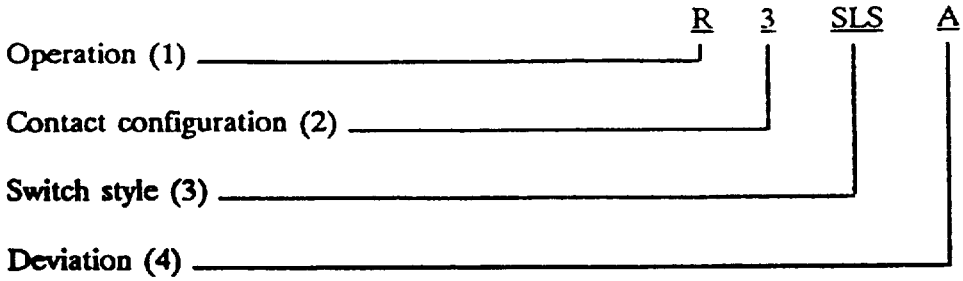
MOUNTING BLOCK POSITION TABLE

TYPE	"A"	"B"	"C"	"D"	CONFIG.
R3SLS	.050	2.800	2.400	.250	
R2SLSB	.250	2.400	2.800	.050	
R2SLSA	.250	2.400	2.800	.250	

NOTES:
1. DIMENSIONS ARE IN INCHES.
2. UNLESS OTHERWISE SPECIFIED, TOLERANCES ARE ±0.01 FOR TWO-PLACE DECIMALS AND ± 0.005 FOR THREE PLACE DECIMALS
3. ARC OF ROTATION TO STOPS. SEE TABLE I FOR SWITCH POSITIONS.

FIGURE 1. *SLS switch assembly*

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(1) Switch assemblies shall be remote operated with manual override, designated R or manually operated, designated M.
(2) Contact configuration shall be type 2 or type 3 as specified (see figures 2 and 3).
(3) Switch style shall be designated by the letters SLS.
(4) Deviation letter is used for type 2SLS switches where two (A) or four (B) switching poles of type 3 contacts are required for motor-control or position indicator circuits.

TABLE I. *Switch selection.*

Type designation	Application data								
	Switch action	Switch poles	Position locations[1]	Load contacts[2]					Control/indicator contacts	
M2SLS	1-OFF-2	40	1 OFF 2	ROW COL	BCD 1-0	EFG 1-0	HJK 1-0	LMN 1-0	ROW COL	– –
M2SLSA	1-OFF-2	36		ROW	BCD	EFG	HJK	LMN	ROW	BCD
M2SLSA	1-OFF-2	36		COL	5-0	1-0	1-0	1-0	COL	1-4
M2SLSB	1-OFF-2	32		ROW	BCD	EFG	HJK	LMN	ROW	BCD
R2SLSB	1-OFF-2	32		COL	9-0	1-0	1-0	1-0	COL	1-8
M3SLS	1-2-3	20	1 – 2 3	ROW COL	BCD 1-0	EFG 1-0	HJK 1-0	LMN 1-0	ROW COL	– –
R3SLS	1-2-3	18		ROW COL	BCD 5-0	EFG 1-0	HJK 1-0	LMN 1-0	ROW COL	BCD 1-4

[1]Position locations are shown relative to the flat on the handle shaft as the switch is viewed from the front (see figure 1) – switches are shown in the center position
[2]Column numbers are inclusive, e g , 1-0 means 1, 2, 3, 4, 5, 6, 7, 8, 9, 0

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REQUIREMENTS:

Construction: Style SLS switch parts, modules, and assemblies shall be constructed in accordance with Drawing 803-5168459.

Characteristics: Characteristics shown in table II shall apply to manual and remote operated switch assemblies.

TABLE II. *Switch characteristics.*

Characteristics	Value
Number of poles	See table I
Number of position	3
Break-before-make	Yes
Make-before-break	No
Electrical rating and endurance loading:	
4 Amperes, 115 V, ac, 60 Hz, p.f. = 1.0	25,000 Operations
2 Amperes, 115 V, ac, 60–400 Hz, p.f = 0.7 (inductive)	25,000 Operations
4 Amperes 28 V, dc, resistive	25,000 Operations
2 Amperes 28 V, dc, inductive, 0.14 henries	25,000 Operations
50 Microamperes – 1.5 V, dc, resistive	25,000 Operations
Endurance shall be run at 5 operations per minute	
Dielectric strength:	
Switch	1000 Vrms at 60 Hz
Motor	500 Vrms at 60 Hz
Insulation resistance:	
Switch	1000 megohm at 500 Vdc
Motor	500 megohm at 500 Vdc
Contact resistance:	
At 4 amperes	10 milliohm
At 50 microamperes	100 milliohm
Temperature rise	30 °C
Current drain at 26 Vdc	2 Amperes
Transfer time at 26 Vdc	5 milliseconds
Actuating time, one position change at 26 Vdc	250 milliseconds
Actuating time, two position change, one direction, 26 Vdc	0.5 second
Temperature:	
Operating	32 °F – 100 °F (0°C – 43°C)
Shock	MIL-S-901, grade A, class I, type C
Vibration	MIL-STD-167-1, type I
Torque – shaft rotation	20 pound-inches
– stop strength	50 pound-inches
Weight	1-3/4 pounds

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Switch assembly:

(a) Manual operated SLS switch assembly shall consist of the following modules as detailed on Drawing 803-5168459.

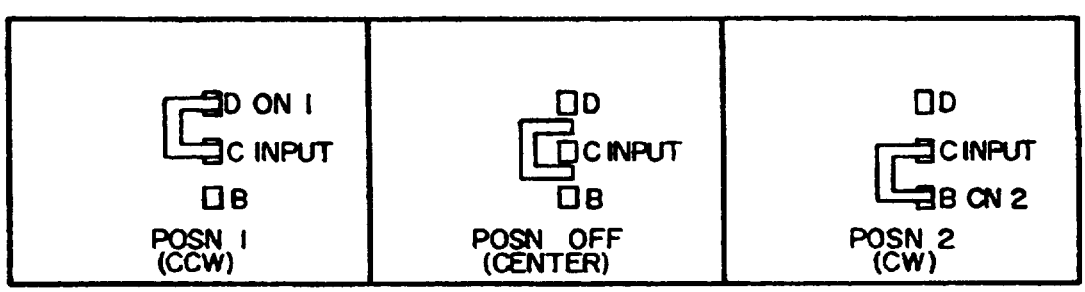
(1) Actuator mechanism assembly
(2) Contact module.

(b) Remote operated SLS switch assembly shall have the following assemblies:

(1) Actuator mechanism assembly
(2) Contact module
(3) Motor assembly
(4) Noise filter.

Switch accessories: A connector plate assembly and a knob assembly (see figure 6) are required for installation and operation of each switch. Accessories to be supplied shall be as specified in the contract or order.

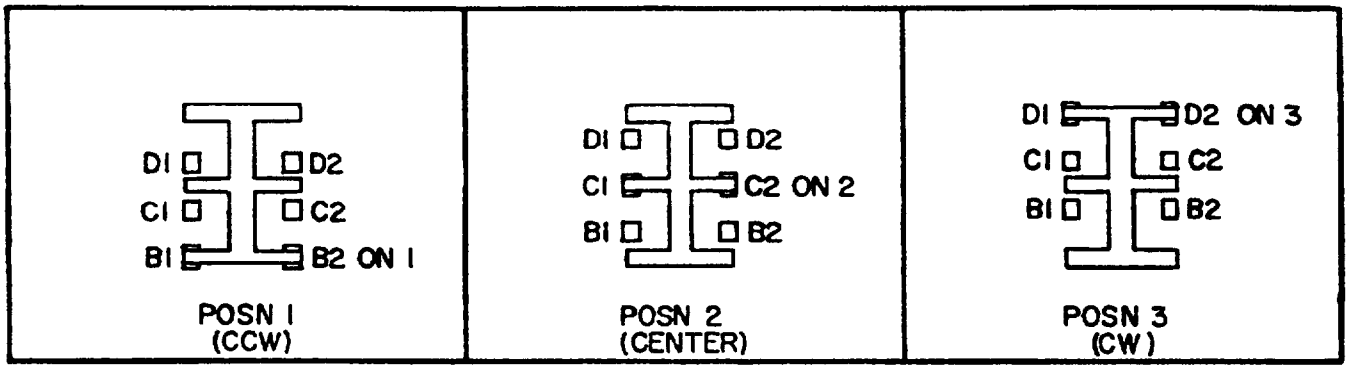
Switch function: Style SLS switch assemblies shall provide up to 40 three-position poles arranged in four groups of three rows of stationary contacts and a movable contact arranged as shown on figures 2 and 3. The rows of stationary contacts are marked B, C, and D, E, F, and G, H, J, and K, and L, M, and N.



NOTE Each pole of a switch having type 2 contacts shall have an "OFF" position and shall be capable of closing a common contact to any one of two separate contacts

FIGURE 2 *Typical operation of a type 2 switch pole*

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NOTE: Each pole of a switch having type 3 contacts shall be capable of closing a common contact to any one of three separate contacts. For a common input within a single pole B1, C1, and D1 may be interconnected.

FIGURE 3. *Typical operation of a type 3 switch pole.*

Contact designation: Contacts of the switch contact module shall be marked as shown on figure 1. When designating contacts in tabular form the sequence for type 2 contacts shall be row letter followed by column numbers, e.g., B2.

Remote operation: Motor circuit for remote operation shall be in accordance with figure 4.

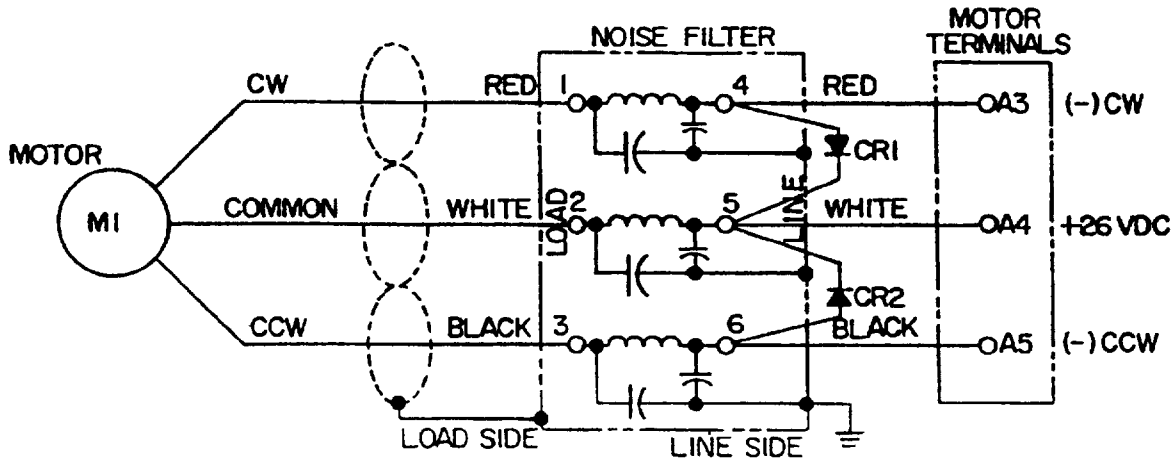


FIGURE 4. *Motor circuit for remote operation*

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NOTE: 26 Vdc applied between terminals A4 and A5 shall drive the moving contacts toward position 1; 26 Vdc applied between terminals A4 and A3 shall drive the contacts toward position 3. For typical control circuit, see figure 4 of MIL-S-24187. When switch contacts move into an adjacent position, power shall be interrupted by the control contacts and the switch shall be held in its commanded position by a mechanical detent. Control of power interruption shall be by means of external interconnections and by using contacts BE-B4, C1-C2, C3-C4, and D1-D2. Contacts B1-B2 and D3-D4 may be used for position monitoring.

CAUTION

WHEN TEMPORARILY OPERATING MOTORIZED SWITCH, APPLY CONTROL VOLTAGE TO MOTOR TERMINALS MOMENTARILY. PERMANENT DAMAGE WILL RESULT IF CONTROL VOLTAGE IS MAINTAINED MORE THAN TWO SECONDS.

FIGURE 4. *Motor circuit for remote operation – Continued.*

Repair parts: Repair parts shall be as specified in table III.

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TABLE III. *Onboard repair parts.*

Drawing 5168459 piece No.	Description	Quantities required for quantity installed					
		1 to 3	4 to 12	13 to 30	31 to 60	61 to 120	121 or more
1	Switch assembly type R3SLS	1	1	1	2	3	4
2	Switch assembly type R2SLSB	1	1	1	2	3	4
3	Switch assembly type R2SLSA	1	1	1	2	3	4
121	Connector plate assembly R3SLS	1	1	1	2	3	4
122	Connector plate assembly R2SLSB	1	1	1	2	3	4
123	Connector plate assembly R2SLSA	1	1	1	2	3	4
100	Switch knob (short) assembly	1	2	3	5	5	10
101	Switch knob (long shaft) assembly	1	2	3	5	5	10
111	Shaft coupling	1	2	3	5	5	10
110	Extension shaft (long shaft)	1	2	3	5	5	10
128	Terminals bushing, 10 position	1	1	1	1	2	2
129	Terminal, crimp	See note					
	Repair parts for field maintenance (field maintenance procedures required)						
200	Motor assembly	0	1	1	2	2	3
5	Noise filter	1	1	1	2	2	3
70	Silicon diode	2	2	4	6	6	10
4	Actuator mechanism assembly	0	1	1	2	2	3
7	Contact module R3SLS	0	1	1	2	2	3
8	Contact module R2SLSB	0	1	1	2	2	3
9	Contact module R2SLSA	0	1	1	2	2	3

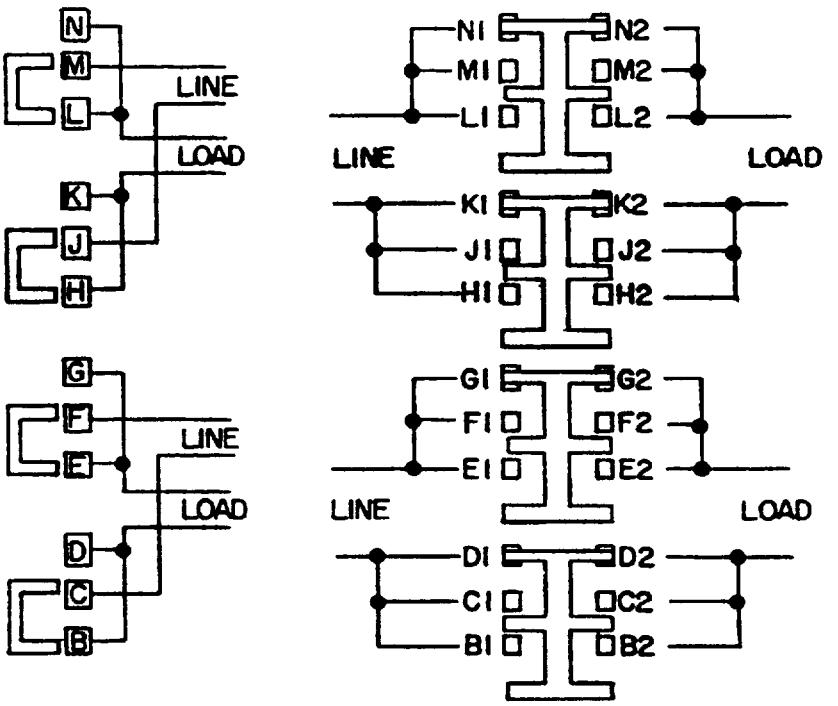
NOTE: Supply 10 plus one for each 100 or fraction of 100 installed.

Qualification samples: Two remote operated switch assemblies shall be submitted for qualification inspection.

TESTS:

Loading of switches: Loading of switches for endurance and temperature rise tests for qualification and quality conformance, group C testing, shall be in accordance with figure 5

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Type 2SLS		Contact load		Load circuit components	Type 3SLS	
Columns	Rows	Load voltage	Load current, type and power factor		Columns	Rows
7, 8, 9, 0	L, M, N	28 V dc	4 amperes, resistive	Noninductive resistor	5-6, 7-8	L, M, N
3, 4, 5, 6	L, M, N	28 V dc	2 amperes, inductive, 0.14 ± 0.05 henries	Air-core	9-0 9-0	H, J, K L, M, N
1, 2	E, F, G	115 V ac, 60 Hz	2 amperes, inductive, 70 percent p.f.	Reactor and resistor	1-2, 3-4	L, M, N
9, 0	B, C, D	115 V ac, 400 Hz	2 amperes, inductive, 70 percent p.f.		5-6, 7-8	B, C, D
1, 2 1, 2	H, J, K L, M, N	115 V ac, 60 Hz	4 amperes, 100 percent p.f	Noninductive resistor	1-2 1-2	E, F, G H, J, K
9, 0 9, 0	E, F, G H, J, K	1.5 V dc	50 micro-amperes	Noninductive resistor	9-0 9-0	B, C, D E, F, G

FIGURE 5 *Test loading*

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Temperature rise: Contacts shall be monitored for temperature rise test as specified in table IV.

TABLE IV. *Temperature measurements.*

Type R2SLS		Type R3SLS	
Column	Row	Column	Row
1 1 9 0	K M F H	1 1 0 0	E H C F

Accessories: Accessories shall consist of those shown on figure 6.

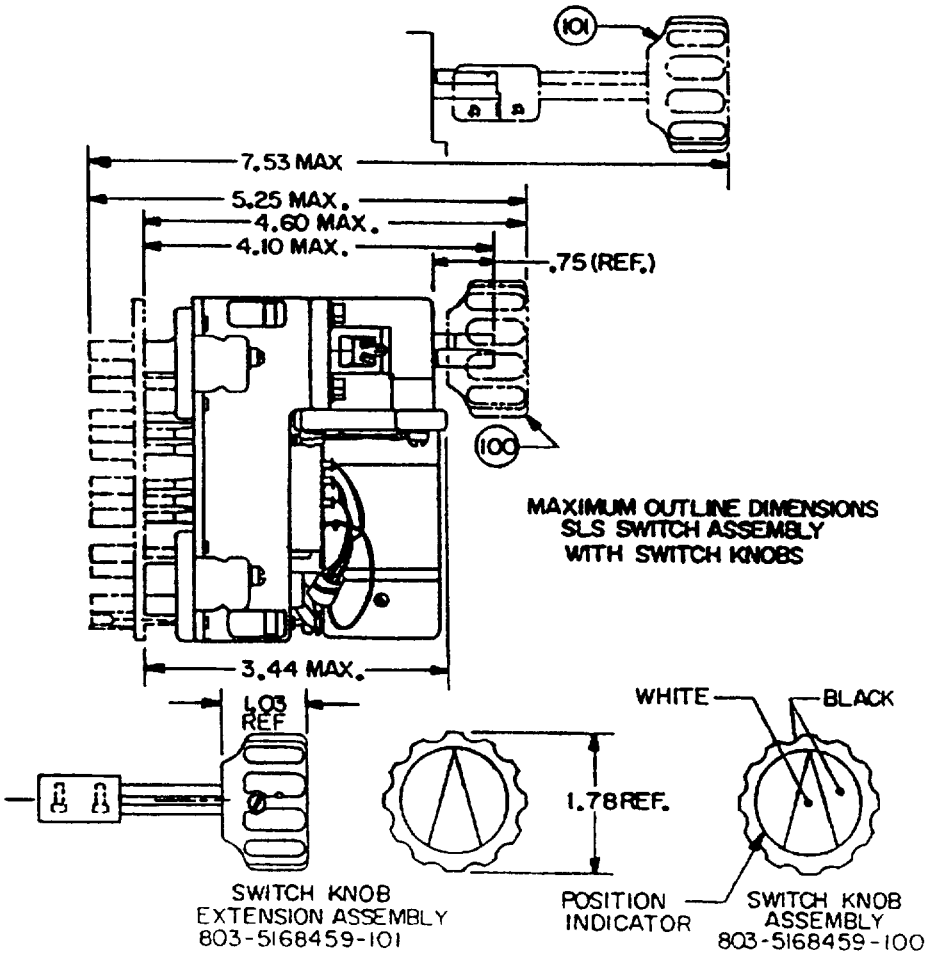
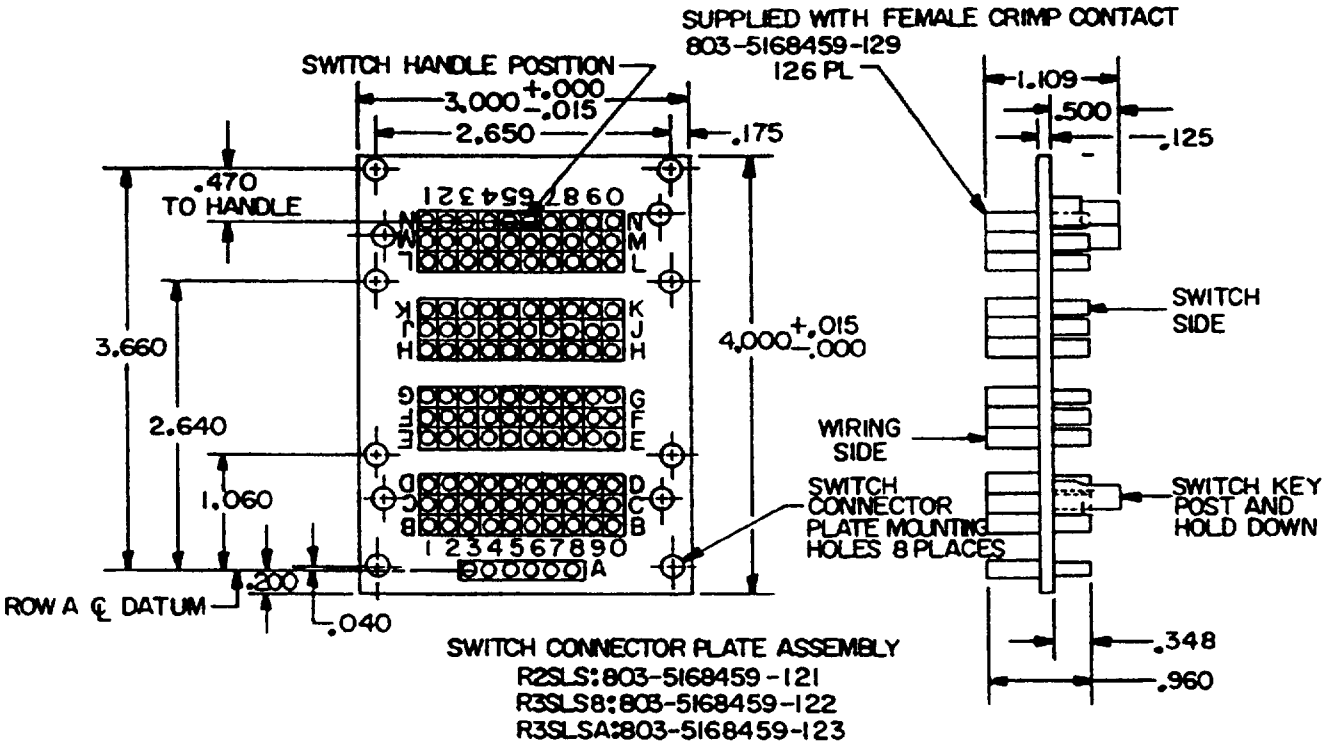


FIGURE 6 *SLS switch accessories*

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NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±0.01 for two-place decimals and ±0.005 for three-place decimals.

FIGURE 6. *SLS switch accessories – Continued.*

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity
Navy – SH
(Project 5930-N681-07)

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